

Predicting TCR Cross-Reactivity Before It Reaches a Patient

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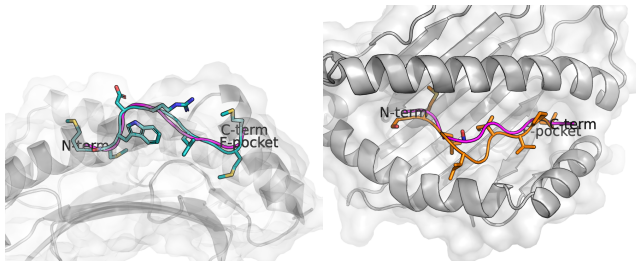


Figure 1: **Proposed framework.** Diffusion models sample from the cognate conformation (teal, left) toward a known off-target one (orange, right). Each generated backbone (magenta) is scored against both.

The problem. TCR-engineering therapies promise a transformative future for cancer treatment, and with the 2024 approval of afamitresgene autoleucel that future has begun to arrive¹. Yet cross-reactivity remains a fundamental barrier to scaling them into personalized medicine. The danger is not hypothetical. An affinity-enhanced MAGE-A3 receptor that had cleared extensive preclinical testing caused fatal cardiac toxicity in two patients², cross-recognizing an unrelated self-peptide from the muscle protein titin that was identified only after the fact³.

Why the field is stuck. The bottleneck has never been imagination, it has been throughput. Experimental validation is slow and costly, an untenable price when the presented peptide space is astronomically large, on the order of 10^{11} , and the fast-growing tumors these therapies target leave only a narrow clinical window to act. The extreme polymorphism of HLA leaves binding data sparse for most alleles and for underrepresented populations, precisely where personalized therapy is needed most. Amino-acid scanning resolves one receptor at a time and does not scale to the combinatorics of the peptide-TCR interaction.

The approach. The tools to test this at scale have only now come within reach. Two model classes make it possible. Diffusion models generate backbones by iteratively denoising random coordinates into a structure consistent with a supplied context. Inverse folding solves the converse problem, encoding each residue's geometric environment as a graph and decoding the sequence most compatible with it, learned without any binding measurement⁴. I would bring two complementary tools of my own design built on them. ESMCBA is a binding-affinity system built on protein language models with domain adaptation that recovers usable signal even for data-scarce MHC alleles⁵. The second uses diffusion models conditioned on the MHC groove to sample compatible pep-

tides across high-priority alleles⁶. Together they let me computationally screen these interactions and chart the cross-reactive neighborhood of a receptor before it ever reaches a patient.

How it is proven. Each prediction is falsifiable at the bench. The receptor either engages the flagged peptide or it does not. The scoring criterion comes first. Holding MHC and receptor fixed and leaving the peptide de novo, generation is conditioned only on receptor-side contacts, sweeping their number to locate the breadth at which a target conformation becomes reachable. Each backbone is scored on three criteria that must agree, proximity to target, occupancy of the B and F anchor pockets, and burial depth, with thresholds calibrated against native molecular-dynamics ensembles of both conformations so the envelope has a defined width rather than an arbitrary cutoff (Fig. 1). Threading direction is checked explicitly, since a reversed chain scores well on deviation alone. In parallel, four inverse-folding models applied to twenty solved pMHC-TCR structures, with the receptor present and deleted, identify the positions whose recovery collapses on deletion, which are the positions that receptor reads. These are validated against measured mutations from SKEMPI and IEDB matched by CDR3, and an architecture-matched variant retrained without MHC and TCR structures separates learned chemistry from memorization. Validation then proceeds in two stages. Retrospectively, the screen runs blind against receptors whose off-targets are documented and must recover them, with MAGE-A3 as the primary test, since the titin epitope is known and was missed by the methods then in use. A screen that fails to flag it fails outright. Prospectively, top-ranked candidates go to binding measurement and, for binders, to functional assay, yielding what the field lacks, a false-negative rate on peptides never seen in training. The loop closes within the appointment.

What becomes possible. A screen that runs in silico before infusion turns cross-reactivity from a lethal surprise into a predictable and testable step, applied to every candidate receptor rather than the few that fit an experimental budget, and making the next generation of cancer therapies measurably safer. It generalizes to any therapy built on engineered molecular recognition, where the failure mode is likewise specificity, not potency. The societal stake is measured in patients, not benchmarks.

Trajectory. The first problem is chosen because it is winnable inside a year, the second because it is worth a career. Near term the deliverable is the validated screen and its false-negative rate. Beyond it lies whether specificity can be designed at the outset, not audited after.

Recent unpublished work of my own indicates both routes are tractable and that the discrimination criterion behaves as intended on real designs, which is what makes this worth building now.

¹ Singh, Mol. Ther. 32:3195, 2024.

² Linette et al., Cardiovascular toxicity and titin cross-reactivity of affinity-enhanced T cells. Blood 122:863–871, 2013.

³ Cameron et al., A titin-derived HLA-A1 peptide as cross-reactive target for MAGE-A3 T cells. Sci. Transl. Med. 5:197ra103, 2013.

⁴ Ribeiro-Filho et al., Structure-based deep learning for TCR design. PLOS Comput. Biol., 2024.

⁵ Mares, Espinoza, Ioannidis, Domain-specific pre-training of protein language models for pMHC-I binding. PMLR 311:304–325, 2025.

⁶ Mares, Espinoza, Ioannidis, Structure-guided pMHC-I libraries using diffusion models. ICML GenAI Biology Wksp, 2025.